LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART IV

SESSION 14

Country Risk & Scenario Planning

From markets, to macro, to decisions that matter — in Nigeria and the world.



❐ Reading: India: Liberalise in the Face of Crisis? (P86) · Facilitator: Sumendra Dash

TODAY'S PLAN

Session 14 at a Glance — 75 Minutes

How to compare countries honestly — and how to plan when you cannot forecast.



Recap & Hook

Where should we expand next?

Country-Risk Scoring

Indicators, z-scores, pillars and weights

Scenarios & Stress Tests

Scenario trees, signposts, tornado charts

Lab: Rank & Stress-Test

Eight markets, one Nigeria plan

BY THE END OF TODAY

Session 14 Learning Objectives

You will be able to:

01

Score Country Risk

Build a transparent country-risk score from political, policy, external, fiscal and banking indicators.

02

Plan with Scenarios

Replace a single forecast with a few plausible scenarios, probabilities and signposts.

03

Stress-Test a Plan

Find the shocks that hurt a business plan most, and decide what to do about them in advance.

THE HOOK

Where Should We Expand Next?

“Nigeria is our biggest market, so it must be our best market.”

Size is only one side of the question. A regional CEO also asks:

How risky is it?

Can we get dollars out? Will policy change overnight? Will the currency halve?

Risky compared with what?

A score means little alone. Rank markets against each other using the same yardstick.

What if we're wrong?

No-one forecast the size of the 2023–24 naira move. Plans must survive surprises.

Eight Markets, One Expansion Budget

The situation

A Lagos-based consumer-goods group will expand into **one more African market** and must also defend its **Nigerian** business, which earns ₦18bn EBITDA on ₦120bn revenue with 45% of costs imported.

The decision question

How do the eight candidate markets **rank on risk**, does the ranking survive different weightings, and how badly could the **Nigeria plan** be hit next year?

The data we have

country_risk_panel.csv — eight African markets × ten indicators: growth, inflation, debt, reserves, current account, governance, FX depreciation, Eurobond spreads. *Stylised teaching data.* Plan model assumptions are stated on each chart.

What a good answer looks like

A **ranked shortlist** with the weighting stated openly, a **three-scenario plan** for Nigeria, and a **signpost board** that tells management when to switch plans.

PART

01

Scoring Country Risk

Make your judgement explicit — then test it.

Five Families of Country Risk

Risk family	The question	Indicators used here
Macro	Is the economy growing with stable prices?	Real growth, inflation
External	Can the country pay for imports and service foreign debt?	Reserves in months of imports, current account, 12-month FX depreciation
Fiscal & market	Can the government pay its bills; what do lenders charge?	Public debt / GDP, Eurobond spread
Political & policy	Will rules stay stable and be enforced?	World Bank governance indicators: political stability, regulatory quality
Banking	Is the financial system sound?	Not scored here — add capital ratios, NPLs, credit growth (Session 10)

Rating agencies, export-credit agencies and banks use richer versions of the same logic.

SAME YARDSTICK

Turning Different Indicators into One Score

IN PLAIN ENGLISH

Inflation is in %, reserves in months and governance on a −2.5 to +2.5 scale. To combine them, express each as a **z-score**: how far a country is from the group average, in standard deviations. Flip the sign where high is good, so **higher always means riskier**.

EXAMPLE

Nigeria's political-stability score (−1.9) is well below the group average, giving a high risk z-score. Its reserves (7 months) are above average, giving a low one.

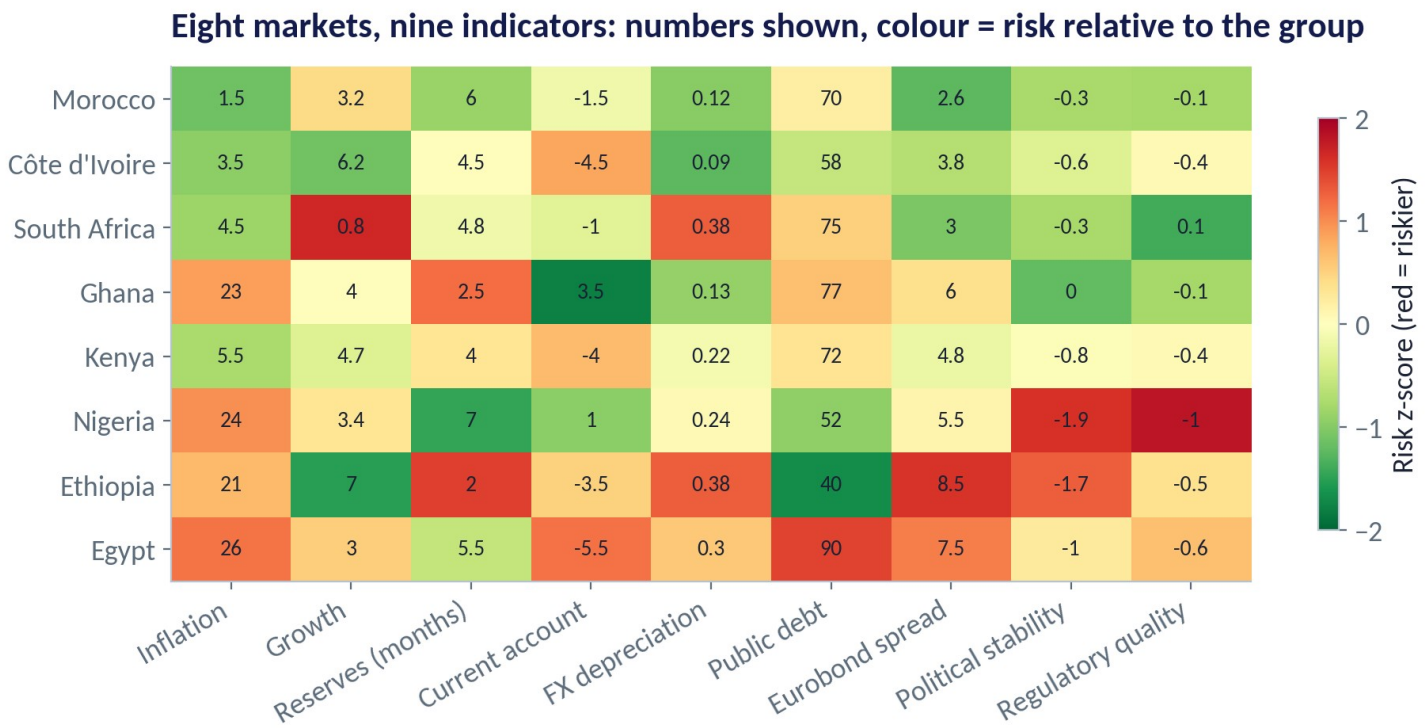
THE FORMULA

$$z = (\text{value} - \text{group average}) \div \text{standard deviation}$$

WHY A MANAGER CARES

Writing the method down makes your judgement **auditable**. Colleagues can argue about the weights instead of arguing about gut feel.

Eight Markets at a Glance



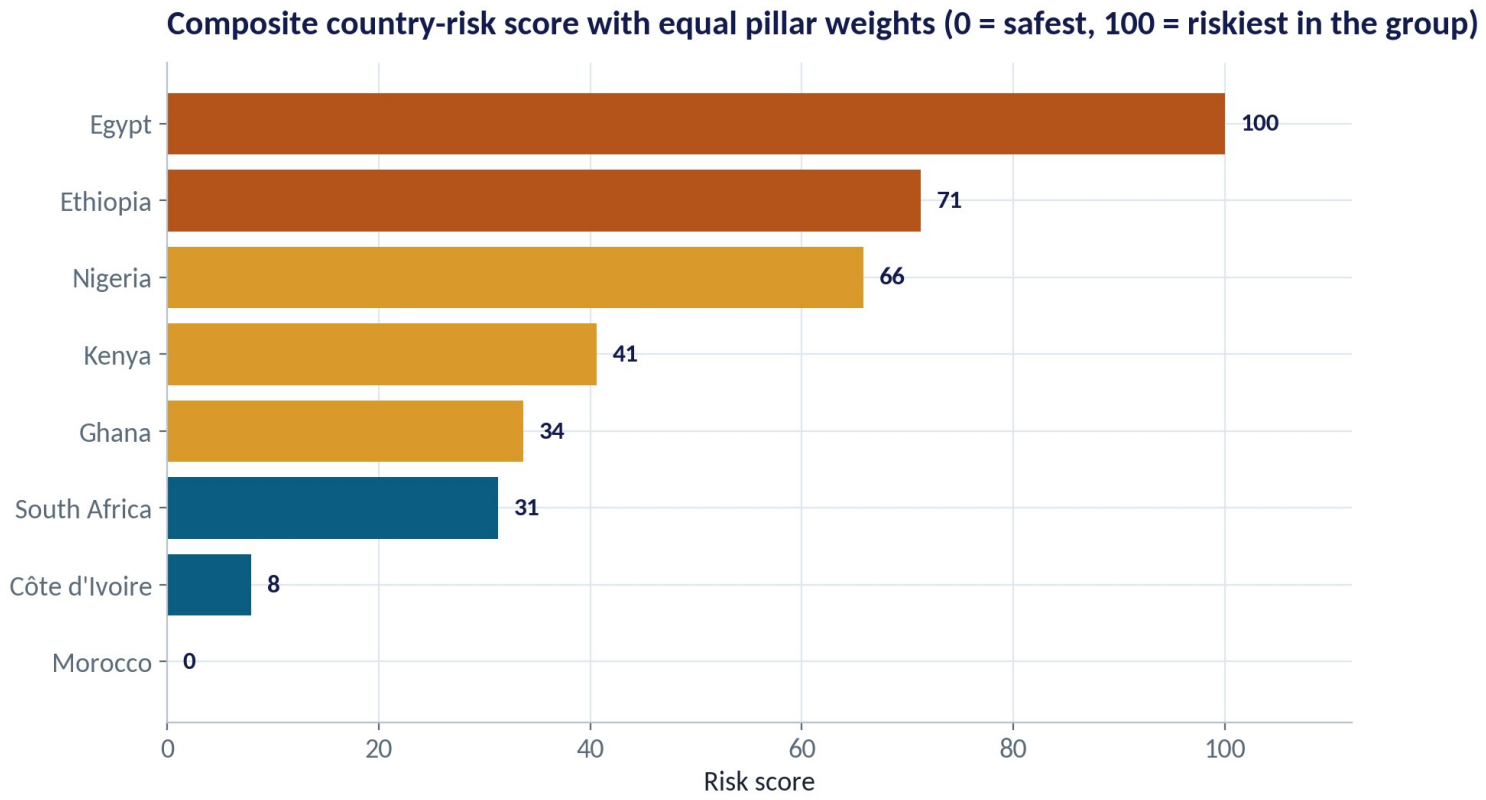
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S14-F01 · replicate: s14_figures.R / .py

WHAT TO SEE

- Rows are markets, columns indicators; the number is the raw value, the colour its relative risk.
- **Morocco** and **Côte d'Ivoire** are green across most columns.
- **Nigeria** is red on governance and inflation, but green on reserves and debt.
- **Egypt** and **Ethiopia** are red on debt, spreads and FX.

Equal Weights: Who Is Riskiest?



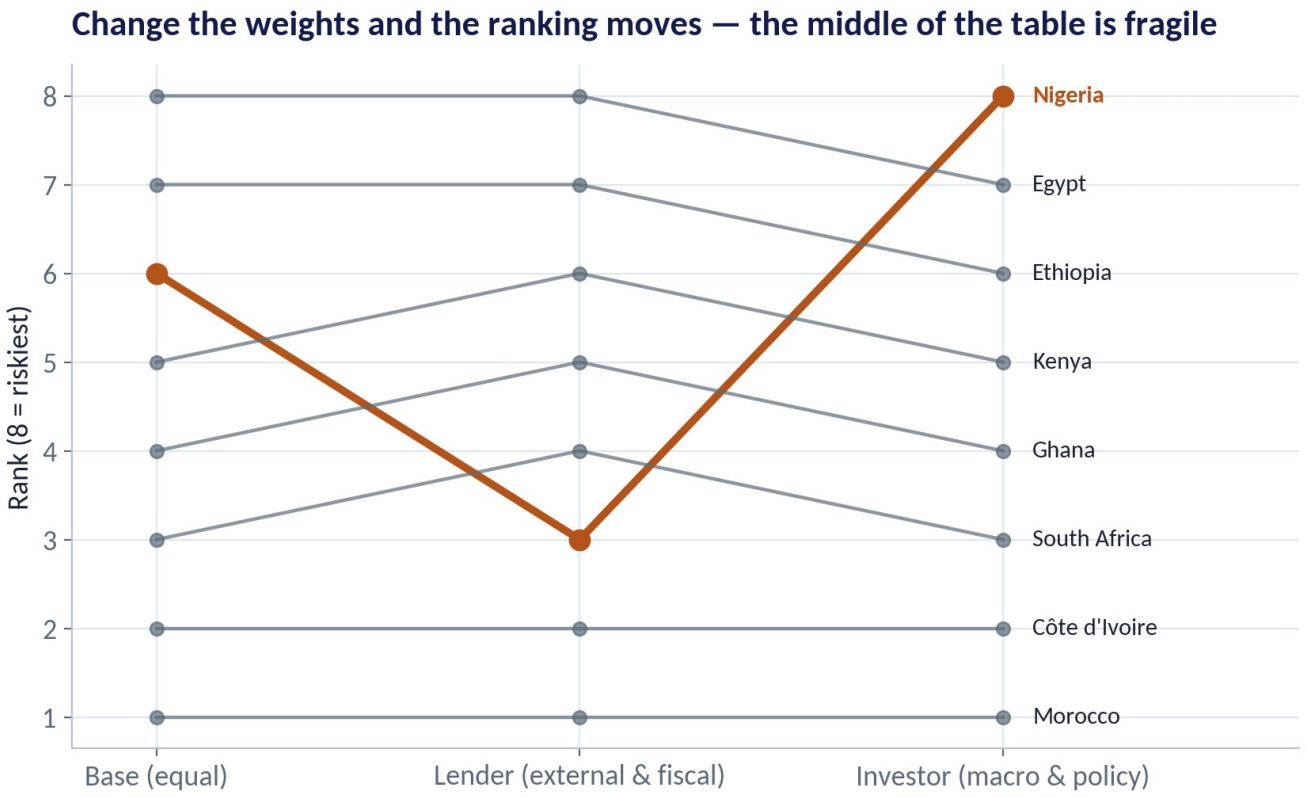
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S14-F02 · replicate: s14_figures.R / .py

WHAT TO SEE

- Averaging the four pillars equally and rescaling to 0–100.
- Morocco (0) and Côte d'Ivoire (8) are the safest in this group.
- **Nigeria scores 66**, behind Ethiopia (71) and Egypt (100).
- A score is **relative** to this group, not an absolute probability of default.

Change the Weights, Change the Answer



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S14-F03 · replicate: s14_figures.R / .py

WHAT TO SEE

- A **lender** cares most about external and fiscal risk: Nigeria rises to **3rd safest** thanks to reserves and moderate debt.
- An **investor** cares most about macro and policy: Nigeria becomes the **riskiest** of the eight.
- The ends of the table are stable; the **middle is fragile**.
- Always report the weights — and test whether your decision changes if they change.

Big Markets Are Not Always Safe Markets



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S14-F04 · replicate: s14_figures.R / .py

WHAT TO SEE

- Vertical axis: market size in five years (population grown at the real growth rate).
- Nigeria, Ethiopia and Egypt are the **big prizes** — and among the riskiest.
- Morocco and Côte d'Ivoire are safer but smaller.
- Strategy is choosing **where on this map** your firm can earn a premium for the risk it takes.

CONCEPT CHECK

Quick Check: What Would You Say?

?

A colleague's spreadsheet ranks Nigeria 3rd safest of eight markets; yours ranks it 8th. Both use the same data. What is the most likely explanation?

A. One of you made an arithmetic error

B. You used different weights: hers stresses reserves and debt, yours stresses inflation and governance

C. Nigeria's data are unreliable

D. Country risk cannot be measured

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Scenarios, Signposts and Stress Tests

Plan for several futures, not one forecast.

TWO WAYS TO PLAN

A Single Forecast vs Scenario Planning

Single-point forecast

"The naira will be ₦1,550 next year."

- One number, false precision
- Plan breaks when the number is wrong
- Invites optimism bias
- Hard to know when to change course

Scenario planning

"Three plausible worlds, with probabilities."

- A few coherent stories (FX, inflation, rates, demand together)
- Actions prepared for each
- Signposts tell you which world you are in
- Probability-weighted value guides investment

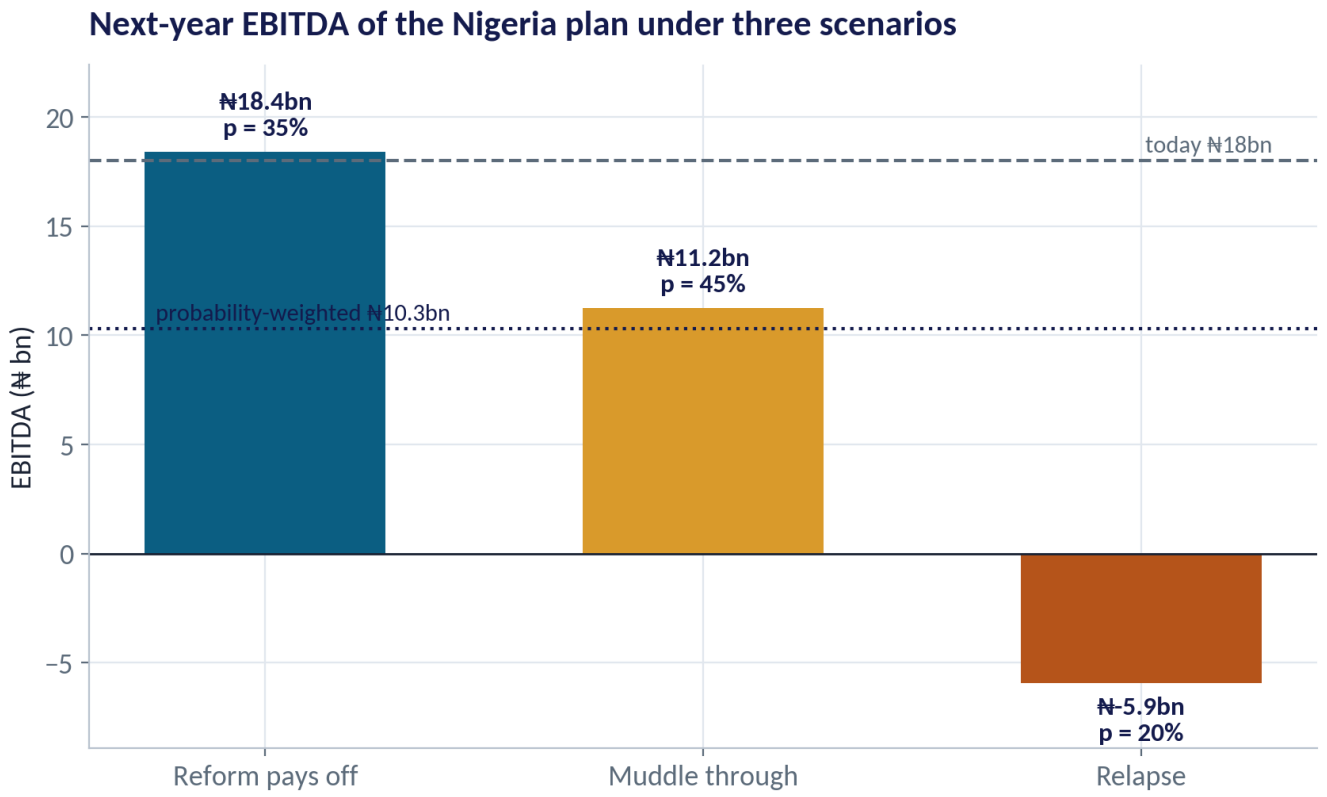
Scenarios are not forecasts. They are a disciplined way to ask 'what would we do if…?' before it happens.

Three Scenarios for Next Year

Scenario	Probability	Naira	Real demand	Price pass-through	Policy rate
Reform pays off	35%	5% weaker	+10%	80%	−3 points
Muddle through	45%	25% weaker	+1%	60%	unchanged
Relapse	20%	80% weaker	−4%	40%	+6 points

Each scenario moves several variables together, the way real economies do. Probabilities are judgement — write down why.

The Plan's EBITDA Under Three Worlds



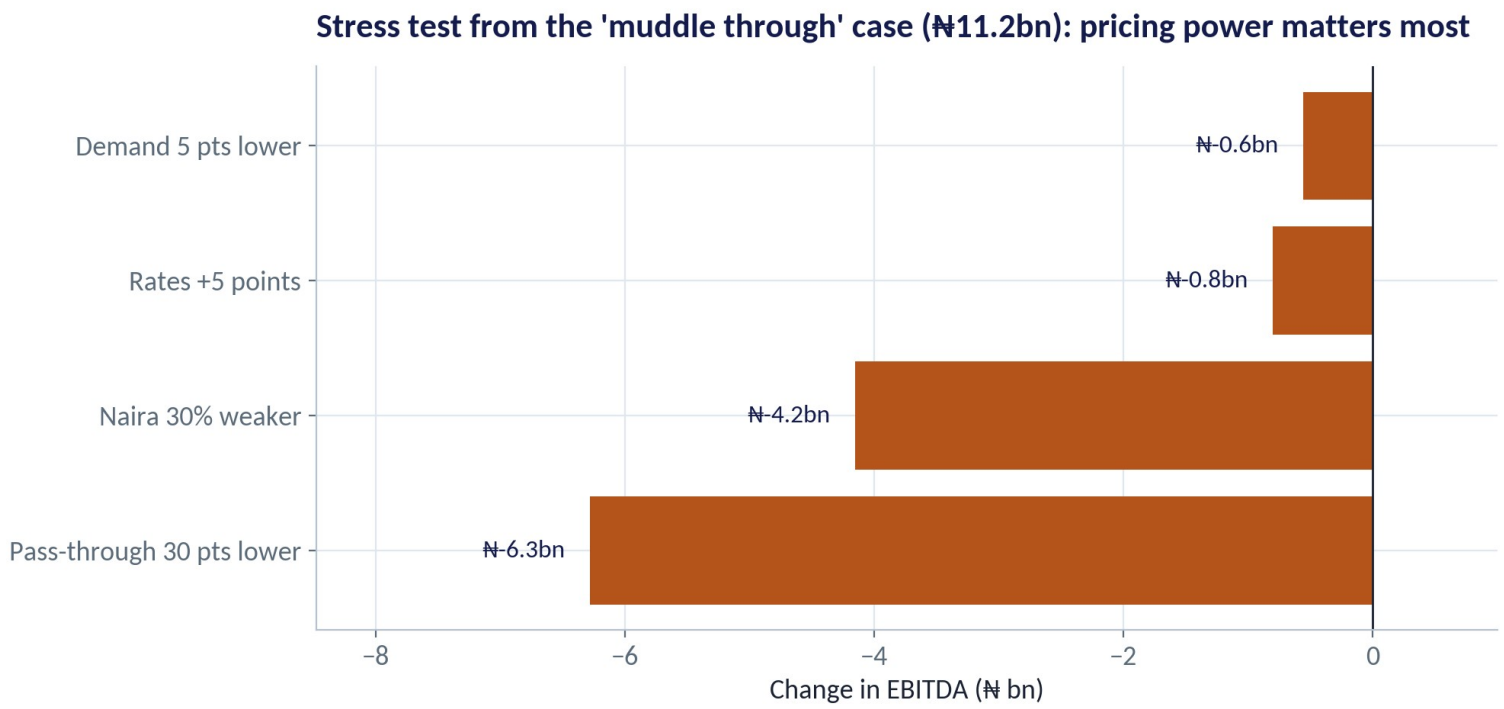
Illustrative plan model: ₦120bn revenue, 45% of costs imported, elasticity −0.8. Scenario values are teaching assumptions.

Fig S14-F05 · replicate: s14_figures.R / .py

WHAT TO SEE

- Today's EBITDA: **₦18bn**.
- Reform: **₦18.4bn**. Muddle through: **₦11.2bn**. Relapse: a **loss of ₦5.9bn**.
- Probability-weighted EBITDA: **₦10.3bn** — well below today.
- The relapse case decides how much debt and fixed cost the plan can carry.

Which Shock Hurts Most?



Illustrative plan model. EEGB teaching assumptions.

Fig S14-F06 · replicate: s14_figures.R / .py

WHAT TO SEE

- Starting from 'muddle through' (₦11.2bn), apply one shock at a time.
- Losing **30 points of pricing power** costs about **₦6.3bn** — the biggest hit.
- A further **30% naira fall** costs about **₦4.2bn**.
- Rates and demand matter less for this firm — so focus hedging and pricing strategy there.

Signposts and Triggers

IN PLAIN ENGLISH

A **signpost** is an indicator that tells you which scenario is unfolding. A **trigger** is the level at which you act — agreed **in advance**, so the decision is not made in panic.

EXAMPLE

If 12-month naira depreciation passes **30%** (red), switch to the relapse playbook: raise prices faster, cut dollar inventory, delay capex.

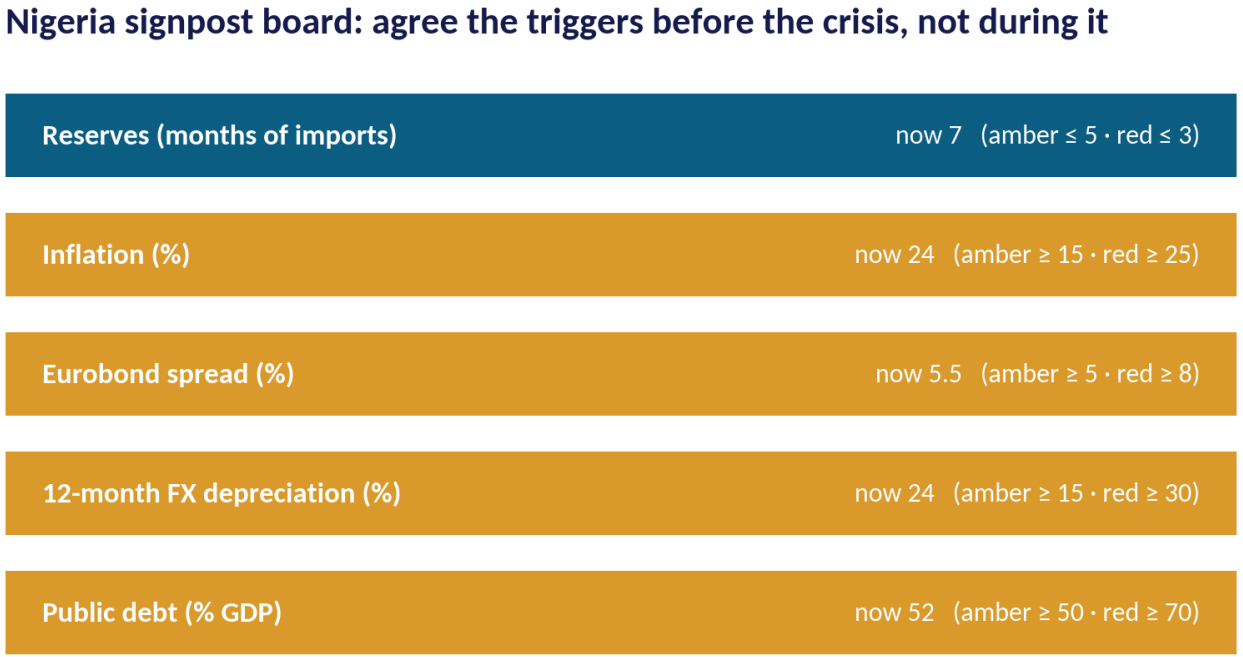
THE FORMULA

signpost crosses trigger → pre-agreed action

WHY A MANAGER CARES

Most firms see the warning signs and still react late. Pre-agreed triggers turn foresight into action.

Nigeria Today: Green, Amber or Red?



Trigger levels are illustrative; set your own with your treasury and risk teams. EEGB stylised teaching data.

Fig S14-F07 · replicate: s14_figures.R / .py

WHAT TO SEE

- Reserves at **7 months** of imports: green.
- Inflation (24%), Eurobond spread (5.5%), FX depreciation (24%) and debt (52% of GDP): **amber**.
- No indicator is red — consistent with 'muddle through'.
- Review monthly with the dashboard from Session 1.

How Country-Risk Analysis Goes Wrong

THE TRAP	THE BETTER HABIT
✗ Ranking countries without stating the weights.	✓ Publish the **weights** and show how the ranking changes.
✗ Using one 'base case' forecast for the budget.	✓ Use **three scenarios** that move FX, inflation, rates and demand together.
✗ Assigning a tiny probability to the bad case so it can be ignored.	✓ Ask what the bad case does to **debt covenants and liquidity**.
✗ Watching dozens of indicators.	✓ Pick **five signposts** with agreed triggers.
✗ Treating a high-risk market as a no-go.	✓ Ask whether the **return** and your **capabilities** justify the risk.

Quick Check: What Would You Say?

?

The stress test shows that losing pricing power hurts the Nigeria plan more than a further 30% naira fall. Which action follows most directly?

A. Buy more dollar forwards

B. Invest in brand strength, product mix and pricing processes so price increases stick

C. Exit Nigeria

D. Ignore the result because the naira is the bigger headline risk

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Rank Eight Markets, Stress-Test Nigeria

YOU WILL NEED

Data
`data/country_risk_panel.csv`

R script
`sessions/s14/R/s14_lab.R`

Python script
`sessions/s14/python/s14_lab.py`

20 minutes

1. Convert the nine indicators into **risk z-scores** (flip the sign where high is good).
2. Average them into **four pillars**.
3. Rank the markets under **equal, lender and investor** weights.
4. Compute the plan's **probability-weighted EBITDA** across three scenarios.
5. Choose **three signposts** and the trigger levels that would change your plan.

Deliverable: a ranked shortlist with your weights stated, and a one-paragraph scenario note for the Nigeria plan.

WORDS TO OWN

Today's Vocabulary in Plain English

Country risk	The chance that conditions in a country hurt an investment or business.
Transfer risk	The risk of not being able to convert or move money out.
z-score	Distance from the group average, in standard deviations.
Pillar	A group of related indicators combined into one sub-score.
Weighting	How much each pillar counts in the final score.
Scenario	A coherent, plausible story about the future.
Scenario tree	Scenarios with probabilities and outcomes laid out together.
Signpost	An indicator showing which scenario is unfolding.
Trigger	A pre-agreed level at which management acts.
Stress test	Measuring the impact of a severe but plausible shock.

KEY TAKEAWAYS

Session 14 in 5 Ideas

01 Country risk has **several families**: macro, external, fiscal, political and banking.

02 Put indicators on **one scale** (z-scores) and make the **weights explicit**.

03 Rankings can **flip** with the weights — test before you decide.

04 Plan with **three scenarios** and a probability-weighted outcome, not one forecast.

05 Agree **signposts and triggers** in advance so action is fast.



NEXT › Session 15: Hedging & Resilience

Before next session: Read Acharya, Almeida, Amihud & Liu, 'Corporate Resiliency and the Choice Between Financial and Operational Hedging' (NBER). Note one example of an operational hedge in your own company.